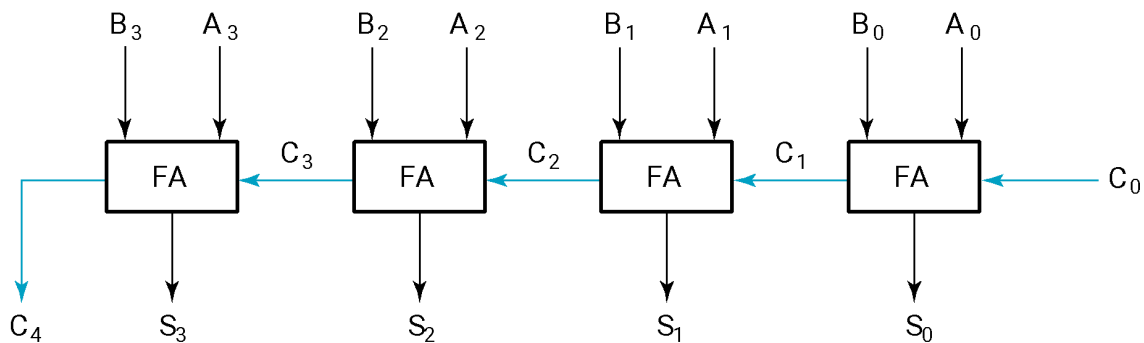




Determine the correct inputs to the full adder for each given arithmetic operation.

**Reminder:** The available inputs are  $X = X_3X_2X_1X_0$  and  $Y = Y_3Y_2Y_1Y_0$ . This means that **any entries in this table should be  $X_i$ ,  $Y_i$ , 0, or 1** (where  $i$  is between 0 and 3). You are using this table to say what you want to connect to the  $A_i$ ,  $B_i$ , and  $C_0$  inputs. It does not make sense to say, for example, that you want to connect  $A_2$  to  $A_2$ . That would mean that the  $A_2$  input is not connected to anything, and has no input.

### PREVIEW: Learning Experience C.1.a

**This is not part of the case study and does not need to be submitted with the case study.**

It is intended to help you get started on LE C.1.a. You will need to follow the same process in order to complete LE C.1.a, starting with the first table. The difference is that the arithmetic operation set is different, and the operands and result are 8-bit vectors, instead of 4-bit vectors.

**Each student/partnership should complete this table as the first step of LE C.1.a.**

Review the Learning Experience C.1.a specification. To get started, select your three elective operations (in addition to the three required operations). Determine the correct inputs to the 8-bit adder for each given arithmetic operation.

**R = Required, C = Choice**

| R or C?  | Operation          | a input to the adder |        |        |        |        |        |        |        | b input to the adder |        |        |        |        |        |        |        | $C_0$ |
|----------|--------------------|----------------------|--------|--------|--------|--------|--------|--------|--------|----------------------|--------|--------|--------|--------|--------|--------|--------|-------|
|          |                    | $A_7$                | $A_6$  | $A_5$  | $A_4$  | $A_3$  | $A_2$  | $A_1$  | $A_0$  | $B_7$                | $B_6$  | $B_5$  | $B_4$  | $B_3$  | $B_2$  | $B_1$  | $B_0$  |       |
| <b>R</b> | <b>add (A+B)</b>   | $A_7$                | $A_6$  | $A_5$  | $A_4$  | $A_3$  | $A_2$  | $A_1$  | $A_0$  | $B_7$                | $B_6$  | $B_5$  | $B_4$  | $B_3$  | $B_2$  | $B_1$  | $B_0$  | 0     |
| <b>R</b> | <b>sub (A-B)</b>   | $A_7$                | $A_6$  | $A_5$  | $A_4$  | $A_3$  | $A_2$  | $A_1$  | $A_0$  | $B'_7$               | $B'_6$ | $B'_5$ | $B'_4$ | $B'_3$ | $B'_2$ | $B'_1$ | $B'_0$ | 1     |
| <b>R</b> | <b>movA (A)</b>    | $A_7$                | $A_6$  | $A_5$  | $A_4$  | $A_3$  | $A_2$  | $A_1$  | $A_0$  | 0                    | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0     |
| <b>C</b> | <b>negA (-A)</b>   | $A'_7$               | $A'_6$ | $A'_5$ | $A'_4$ | $A'_3$ | $A'_2$ | $A'_1$ | $A'_0$ | 0                    | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 1     |
| <b>C</b> | <b>negB (-B)</b>   | 0                    | 0      | 0      | 0      | 0      | 0      | 0      | 0      | $B'_7$               | $B'_6$ | $B'_5$ | $B'_4$ | $B'_3$ | $B'_2$ | $B'_1$ | $B'_0$ | 1     |
| <b>C</b> | <b>subBA (B-A)</b> | $A'_7$               | $A'_6$ | $A'_5$ | $A'_4$ | $A'_3$ | $A'_2$ | $A'_1$ | $A'_0$ | $B_7$                | $B_6$  | $B_5$  | $B_4$  | $B_3$  | $B_2$  | $B_1$  | $B_0$  | 1     |